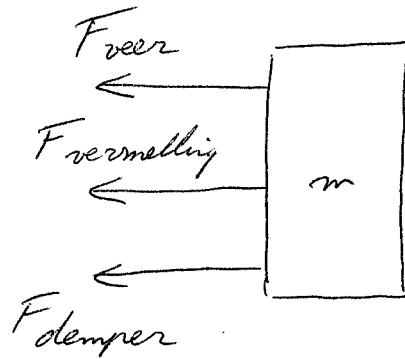


§ 2.8 Opdraven

2a.



$$\Sigma F = 0$$

$$F_{\text{veer}} = c_v (y(t) - x(t)) = c_v \cdot y(t) - c_v \cdot x(t)$$

$$F_{\text{versmelling}} = m \cdot a(t) = m \cdot \frac{d^2 y(t)}{dt^2}$$

$$\begin{aligned} F_{\text{dempers}} &= c_w \cdot v(t) = c_w \cdot \frac{d(y(t) - x(t))}{dt} \\ &= c_w \cdot \frac{dy(t)}{dt} - c_w \cdot \frac{dx(t)}{dt} \end{aligned}$$

$$-c_v \cdot y(t) + c_v \cdot x(t) - m \cdot \frac{d^2 y(t)}{dt^2} - c_w \cdot \frac{dy(t)}{dt} + c_w \cdot \frac{dx(t)}{dt} = 0$$

$$-c_v \cdot y(t) - m \cdot \frac{d^2 y(t)}{dt^2} - c_w \cdot \frac{dy(t)}{dt} = -c_w \cdot \frac{dx(t)}{dt} - c_v \cdot x(t)$$

$$m \cdot \frac{d^2 y(t)}{dt^2} + c_w \cdot \frac{dy(t)}{dt} + c_v \cdot y(t) = c_w \cdot \frac{dx(t)}{dt} + c_v \cdot x(t)$$

$$b) \quad m s^2 y(s) + c_w s \cdot y(s) + c_v \cdot y(s) = c_w \cdot s \cdot X(s) + c_v \cdot X(s)$$

$$(m s^2 + c_w s + c_v) y(s) = (c_w s + c_v) \cdot X(s)$$

$$\frac{y(s)}{X(s)} = \frac{c_w s + c_v}{m s^2 + c_w s + c_v}$$